Gawtam Chithra Ramesh

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Education

KTH Royal Institute of Technology Aug. 2024 – Jul. 2026
Master of Science in Systems, Controls, and Robotics - 4.56/5.0 *Stockholm, Sweden*

Indian Institute of Technology Madras Aug. 2019 – Jul. 2024
Dual Degree (B.Tech in Engineering Design and M.Tech in Robotics) *Chennai, India*

Professional Experience

Masters Thesis (Ongoing) Jan 2026 - Jun 2026
ABB Robotics, Manager - Matthew Lock and Jonathan Styrud *Västerås, Sweden*

- Using generative flow matching for safe, compositional translation from natural language to control policies under Signal Temporal Logic constraints for long horizon industrial manipulation.

Device Research Intern on 3D Scene Graph Jul 2025 - Dec 2025
Ericsson Research, Manager - Puren Guler and Hector Caltenco *Lund, Sweden*

- Developed a robust 3DSG pipeline by integrating instance segmentation and tracking into a SOTA framework (Hydra)
- Designed a synchronized RGB-D inpainting pipeline to remove privacy-sensitive objects from 3DSG in realtime.
- Analyzed 3DSG frameworks identifying key research gaps in object merging, parameter sensitivity, and privacy protection
- Reduced sensitivity to parameter tuning, resulting in robust scene interpretation across diverse environments.
- Benchmarked SOTA inpainting algorithms to balance reconstruction fidelity with real-time performance.
- Lead author on research papers submitted to peer-reviewed venues, ICPR26(under-review) and IMPROVE26(accepted)

Graduate Robotics Intern on Modular Collaborative Robots Dec 2022 - Jul 2023
Systemantics India Pvt. Ltd., Manager - Jagannath Raju *Bangalore, India*

- Developed backward-chained Behavior Trees for 6DOF manipulator to enable robust, long-horizon trajectory planning
- Designed and implemented a 6-DOF trajectory generation and control pipeline in C++, achieving real-time joint actuation via low-latency communication with motor controllers using D-Bus and SocketCAN.

Tested Multi Object Detection and Tracking Models June 2021 – July 2021
Blurgs Research Labs Pvt. Ltd. *Remote, India*

- Analyzed 5 State of the art Multi object tracking networks for startup's first product for The Indian Navy
- Trained and tested various MOT networks like **Siam-MOT** and **FairMOT** for Surveillance with Drones and CCTV.
- Used **FairMOT** to track and get the trail of multiple moving objects and Re-identify objects in real-time.

Robotics Intern on Pallet Handling June 2021 – July 2021
Yaskawa Pvt. Ltd, Mentor - Manju Tiwari *Gurgaon, India*

- Modeled and analyzed 3D palletizing algorithm and mechanics for mixed carton sizes using MotoSim and teach pendant.
- Developed a C++ algorithm to effectively use robot's memory when sending data to the robot from computer.

Imaginate Software Labs Pvt. Ltd. May 2020 – August 2020
Software Development Intern *Remote, India*

- Developed a Unity3D toolkit in C# to upload any 3D model in 60+ file formats into a VR-Environment using Autodesk Forge APIs in real-time.
- Developed a basic Heroku app to upload and view the 3D content in the web browser.
- This toolkit acts handy for the users to upload 3D content to the VR environment, without any reliance on engineers.

Battery Pack Design for Hyperloop Pods Oct 2020 – June 2021
Mentored by Prof. Satya Chakravarthy and Prof. TM Muruganandam *IIT Madras, India*

- Designed a batter pack with high discharge rate for higher speeds while ensuring safe power supply to the pod
- Extended battery pack life by 10% by reducing peak operational temperature by 10°C using an Al heat sink.
- Selected in **top 24** teams for European Hyperloop Week held at Valencia, Spain

Graduate Teaching Assistant Aug 2023 – Nov 2023
Field and Service Robotics (ID4060), Mentor - Bijo Sebastian *IIT Madras, India*

- Involved in design and evaluation of assignments based on ROS and CoppeliaSim for 40 students.
- Held meetings with small groups of students with diverse backgrounds to review their assignments and projects.

Research Experience

Graduate Robotics Research on Deformable Object Manipulation Jan 2025 - Nov 2025
Robotics Perception and Learning, KTH, Advisor - Florian T. Pokorny *Stockholm, Sweden*

- Integrated taxonomy-guided vision-language models (Gemini, Qwen) to interpret DOM scenes and map motion primitives
- Built prompt–taxonomy framework to generate structured codes and textual justifications for robot-parsable perception
- Built prompt–taxonomy framework to generate structured, robot-parsable specifications from high-level prompts, demonstrating a pipeline for translating intuitive instructions into formal outputs.
- Performed quantitative analysis to identify failure modes and guide iterative improvements to the taxonomy and prompts
- Currently integrating multi-modal data and temporal cues to enhance the perception pipeline's robustness.
- Authored and submitted the results at ROMADO, an **IROS2025** workshop.

Dual Degree Project on Structure from Motion Jan 2024 - May 2024
INSPIRE Lab, Advisor - Nirav Partel *IIT Madras, India*

- Developed a 3D perception system using COLMAP (Structure from Motion) from monocular RGB images to build cost-efficient human torso reconstruction for more accessible ultrasound.
- Developed an autonomous ultrasound scanning system using ROS, integrating a 5-DoF robotic arm with a single RGB camera to achieve cost-effective 3D perception for medical imaging.
- Implemented motion planning and control using ROS and MoveIt, and simulated the 5DOF manipulation in Gazebo.
- Integrated perception algorithms within ROS, processing 3D point clouds from RGB images to enable accurate trajectory planning for the ultrasound probe.

Offline And Online Motion Planning & Control of Unmanned Aerial Vehicles Sep 2021 - Apr 2022
Young Research Fellowship, Advisor - *Prof. Satadal Ghosh* *IIT Madras, India*

- Developed a Target Driven Motion Planning and Obstacle Avoidance algorithm for UAVs.
- Used Proportional Navigation and Acceleration Velocity obstacles in 2D field to achieve 20% faster and more computer efficient algorithm compared classical aerospace algorithms.

Scholastic Achievements

- Awarded (among 30 of 800+) the IIT Madras Young Research Fellowship based on remarkable research potential.
- Secured 97.64 Percentile in JEE Advanced & 99.10 Percentile in JEE Mains out of 1.5 Million across India.
- Secured an All India Rank of 52 out of 50k in Undergraduate Common Entrance Examination for Design.

Course Projects

Deep Generative Modelling | DD2610 - KTH Mar 2021 - Jun 2021

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Deep Learning Advanced | DD2610 - KTH Mar 2021 - Jun 2021

- Probabilistic
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Robot Learning and Embodied AI | DD2601 - KTH Sep 2025 - Oct 2025

- Implemented a 3D semantic object mapping pipeline using RGB-D data from ARKitScenes, integrating OwlV2 for open-vocabulary object detection, SAM for segmentation, and CLIP for semantic grounding.
- Built a 3D semantic mapping pipeline using RGB-D data and foundation models (OwlV2, SAM, CLIP) for open-vocabulary scene understanding.
- Implemented 2D-to-3D projection with depth fusion and camera geometry to construct dense, queryable semantic maps.
- Implemented 2D-to-3D projection using camera intrinsics and depth fusion to construct dense semantic scene maps with text-based query capability.

- Deployed a pre-trained Vision-Language-Action (VLA) model for robot manipulation, executing policy inference from visual inputs and natural language commands.
- Analyzed model behavior on unseen tasks to study generalization and robustness limits of current VLA architectures.

Multi Agent Path Finding in Unity3D | DD2348 - KTH — Feb 2025 - May 2025

- Developed Hybrid A* path planner and waypoint tracking system in Unity3D for kinematically constrained single-agent (car/drone) navigation, achieving competitive performance.
- Engineered Conflict-Based Search (CBS) to solve MAPF for over 20 holonomic (drones) and non-holonomic (cars) agents, minimizing makespan in shared environments.
- Implemented a greedy dynamic goal assignment strategy for multi-agent teams ("bakery problem") integrated with CBS, optimizing task allocation and minimizing overall travel time.
- battle snake

Reinforcement Learning for Pacman Capture The Flag | DD2348 - KTH — Mar 2021 - Jun 2021

- Designed a hybrid AI for Pacman CTF (Unity3D), merging a custom Behavior Tree for high level strategy with RL models for specialized actions (attack, defend, escape).
- Trained RL (PPO) policies via Unity ML-Agents with role-specific designs (observations, rewards) and advanced methods (phased training, self-play, curriculum learning) for adaptive agent behavior.
- Evaluated strategic agent decision-making and adaptive behavior switching in a competitive, partially observable, dynamic multi-agent Pacman environment.

Control Theory Advanced and Practice | DD2348 - KTH — Mar 2021 - Jun 2021

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Image analysis and Computer vision | DD2348 - KTH — Oct 2024 - Nov 2024

- Applied Fourier Transforms for image filtering and frequency analysis; designed Gaussian smoothing filters via FFT to reduce noise and study effects on varying image resolutions.
- Implemented a multi-scale differential geometry edge detector (computing L_{vv}, L_{vvv} for zerocrossings) and used Hough Transform for line detection from edges.
- Implemented image matching (SIFT, RANSAC) to estimate homographies (planar scenes) and fundamental matrices, enabling 3D scene reconstruction via triangulation

Applied Estimation | DD2348 - KTH — Mar 2021 - Jun 2021

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Introduction to robotics | DD2348 - KTH — Sep 2024 - Oct 2024

- Implemented an iterative inverse kinematics solver from scratch for a 7-DOF KUKA robot arm in ROS
- Applied RRT* to find collision-free paths for a Dubins car, showcasing path planning for non-holonomic vehicles.
- Engineered 2D occupancy grid mapping solutions in ROS, processing laser scans to build maps, identify free space, and generate C-spaces for navigation
- Integrated a Behavior Tree mission planner for a TIAGo mobile manipulator (using ROS and Gazebo), enabling autonomous navigation, vision-based grasping, and robust task execution (e.g., kidnapped robot recovery).

Control of Automotive Systems — Jun 2022 - Nov 2022

Mentored by Prof.Srikanthan Sridharan — *IIT Madras, India*

Control of Automotive Systems | ED—- - IIT Madras — Jun 2022 - Nov 2022

- Designing a heading angle controller for autonomous ground vehicle considering its vehicle dynamics & reduced error between the desired heading angle and the actual heading angle while working within given performance specifications
- Improved performance of pneumatic brake system by designing P and PI controllers to meet given specifications.
- Designed a nonlinear controller (Sliding Mode Control) to attenuate the effect of external forces on a tractor's hydraulic hitch, to reduce excessive vibrations and to prevent front-wheel lift-off.

Human Powered Segway | ED4060 - IITM — Mar 2021 - Jun 2021

- Developed a novel idea of a hybrid between tricycle and segway.
- Deigned a CAD model using Fusion360 and performed stress analysis in COMSOL

Path Finder and Visualizer using PyQt5 — Jun 2021

Mentored by Prof.Ramanathan — *IIT Madras, India*

Path Finder and Visualizer using PyQt5 | ED—- - IITM — Jun 2021

- Developed a 2D Path finder completely from scratch in Python using PyQt5
- The project explores A-Star and Dijkstra Algorithms for Path Finding
- Developed an interactive UI for visualization using QT Designer

Projects

Virtual Reality with Reinforcement Learning shooter Game | CFI, IIT Madras May 2020 - Aug 2020

- Worked on Multi-Agent and Cooperative play using Reinforcement Learning (PPO), simulating a Shooter Strategy Game
- Simulated the agent in Unity3D environment with obstacles & mazes exploring different strategies
- Achieved Superhuman Game Play, used Unity VR to directly implement Virtual Reality through Unity

Python API for Fusion360 | Prof. Ramanathan Jan 2020 - Feb 2020

- Developed a Python script using Fusion 360 API to automate the process of duplicating a 3D model with different dimensions imported from a .csv file
- Eliminates the effort needed to create multiple models manually from scratch for just varying the dimensions.

Discord Bot | *C#* Apr 2021

- Developed a simple bot in C# which can reply and react to texts and reactions.
- The bot also searches for YouTube Videos and converts them to mp3 using ffmpeg and plays them to the user

Technical Skills

- **Languages** : Python, C, C++, C#
- **Robotics** : ROS2, Unity3D, MATLAB, MuJoCo
- **DL** : Pytorch, OpenCV, wandb, JAX
- **Tools** : VS Code, Docker, Git, LaTeX, CUDA, Slurm
- **CAD** : Fusion360, Solidworks, Ansys, COMSOL

Relevant Coursework

Robotics - Introduction to Field and Service Robotics, Mechanics of Serial Robots,
Controls - Control Systems, Controls of Automotive Systems
Programming - Image Analysis and Computer Vision, Fundamentals of Deep Learning, Data Structures and Algorithms

Position of Responsibilities

Student Buddy | THS International Winter Reception 2025, KTH Jan 2025 – Feb 2025

- Co-organized multiple fun games and events to welcome international students at KTH.
- Acted as the primary point of contact for 20 international students part of the buddy group.

Head of Industrial and Public Relations | Dept. of Engineering Design, IIT Madras Apr 2021 – May 2022

- Headed and mentored a three-tier team overseeing placement and internship processes for over 1,200 students
- Actively involved in Alumni and Company Outreach to further improve the opportunities available for students
- Planning & executing the logistic framework for the first ever shift to hybrid mode of the recruitment process

Strategist | Computer Vision and Intelligence Group, IIT Madras Apr 2020 – May 2021

- Responsible for the Budgeting, Managing and Organizing of all CVIG Sessions, GitHub, Instagram, Blog & Discord
- Co-Mentored student projects involving Image Analysis & Reinforcement learning for games
- Co-Organized a Deep Learning Summer School for over 500 students on topics from DL, RL, and CV

Associate Manager | Entrepreneurship Cell, IIT Madras Apr 2020 – May 2021

- Organized Summer Internfair 2020, handling logistics, interviews and congregating 40 startups
- Orgainzed Elevate, a fund-raising competition for startups, bringing over 150 startups from all over the country.
- Managed the operational requirements of board members of unicorns throughout the Annual E-Summit

Extra - Curricular Activities

- Represented IIT Madras Volleyball team in Agrata sports festival Oct'19
- Won National level drawing competition conducted by Times of India Mar'15
- Love playing video games and game development; Reached a rank of Silver 2 in Valorant.